

Start writing your exam paper...

1.	What is the correct negation of the statement: 'All teachers are female'? (A) No teacher is female. (B) Some teachers are not female. (C) All teachers are not female. (D) Some teachers are female.	1 M
2.	What is the value of $\cos 0^\circ$? (A) 0 (B) 1 (C) Undefined (D) 0.5	1 M
3.	If the pair of linear equations $x + y = 10$ and $x - y = 4$ represent the number of girls (x) and boys (y), how many girls are there? (A) 3 (B) 7 (C) 10 (D) 4	1 M
4.	Which of the following statements correctly defines similar figures? (A) They have the same size but not necessarily the same shape. (B) They have the same shape and the same size. (C) They have the same shape but not necessarily the same size. (D) They have different shapes and sizes.	1 M
5.	What is the angle called that is formed by the line of sight with the horizontal when the point being viewed is above the horizontal level? (A) Angle of depression (B) Angle of elevation (C) Angle of incidence (D) Angle of reflection	1 M
6.	Which of the following is NOT a reason why mathematical modelling is important? (A) To gain understanding of a system (B) To predict or forecast future events (C) To memorize exact numbers without approximation (D) To estimate large values that are hard to measure directly	1 M

SECTION A - OBJECTIVE QUESTIONS
(28 Questions = 60 Marks)

7.	What is the HCF of $22 \times 5 \times 7$ and $22 \times 3 \times 13$? (A) 22 (B) 13 (C) 2 (D) 7	1 M
----	---	-----

8.	Which of the following equations is a quadratic equation? (A) $x^2 + 5x + 6 = 0$ (C) $x^3 - 3x + 2 = 0$ (B) $x + 5 = 0$ (D) $2x + 3 = 7$	1 M
9.	One of the broad guidelines followed in the creation of the Class X Mathematics textbook is: (A) Using complex language to encourage higher thinking (C) Including diverse solutions to a problem to encourage creativity (B) Presenting mathematical proofs only in a didactic way (D) Excluding geometric constructions from the textbook	1 M
10.	If the class interval is 25 - 40, what is the class mark used to represent this class for calculating the mean of grouped data? (A) 30 (C) 27.5 (B) 32.5 (D) 25	1 M
11.	Which of the following sequences is an Arithmetic Progression (AP)? (A) 1, 3, 6, 10, 15 (C) 2, 4, 8, 16, 32 (B) 5, 10, 15, 20, 25 (D) 3, 5, 5, 7, 9	1 M
12.	What is the probability of getting a tail when a fair coin is tossed once? (A) $\frac{1}{2}$ (C) 0 (B) 1 (D) $\frac{1}{4}$	1 M
13.	If two lines representing a pair of linear equations in two variables intersect, what kind of solution do these equations have? (A) No solution (C) Infinite solutions (B) Unique solution (D) Dependent solution	1 M
14.	If $p(x) = 3x + 6$, what is the zero of the polynomial? (A) -2 (C) -6 (B) 2 (D) 6	1 M
15.	What is the formula to calculate the area of a sector of a circle with radius r and central angle θ in degrees? (A) $\text{Area} = (\theta / 360) \times \pi \times r^2$ (C) $\text{Area} = 2 \times \pi \times r \times \theta$ (B) $\text{Area} = (360 / \theta) \times \pi \times r^2$ (D) $\text{Area} = \pi \times r^2 / \theta$	1 M

16.	Which Fundamental Duty, according to the Indian Constitution, was inserted by the Constitution (86th Amendment) Act, 2002? (A) To protect and improve the natural environment (B) To provide education opportunities to children between the ages of six and fourteen (C) To uphold and protect the sovereignty of India (D) To cherish and follow noble ideals of the national struggle	1	M
17.	Which of the following represents the coordinates of a point dividing the line segment joining A(x1, y1) and B(x2, y2) internally in the ratio 1 : 2? (A) $((x1 + 2x2)/3, (y1 + 2y2)/3)$ (B) $((2x1 + x2)/3, (2y1 + y2)/3)$ (C) $((x1 + x2)/2, (y1 + y2)/2)$ (D) $((x2 - x1)/3, (y2 - y1)/3)$	1	M
18.	What is the angle called when the line of sight forms an angle with the horizontal looking downwards from an observer to a point below the horizontal? (A) Angle of elevation (B) Angle of depression (C) Vertical angle (D) Complementary angle	1	M
19.	Assertion (A): If a prime number p divides a^2, then p also divides a. Reason (R): The Fundamental Theorem of Arithmetic ensures unique prime factorisation, so the prime p must appear in the factorisation of a if it divides a^2. (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A). (C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.	1	M
20.	Assertion (A): If a pair of linear equations have ratios of coefficients $a1/a2 = b1/b2 = c1/c2$, then they have infinitely many solutions. Reason (R): This occurs because the lines represented by the equations are coincident. (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A). (C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.	1	M

21.	From a point Q, the length of the tangent to a circle is 24 cm and the distance of Q from the centre is 25 cm. Find the radius of the circle.	2	M
22.	Find the roots of the quadratic equation $2x^2 - 5x + 3 = 0$ by factorisation.	2	M
23.	Find the distance between the points (2, 3) and (4, 1) using the distance formula. OR Find the zeroes of the quadratic polynomial $p(x) = x^2 - 5x + 6$ and verify the sum of its zeroes.	2	M
24.	The probability of Sangeeta winning a tennis match is 0.62. What is the probability of Reshma winning the match?	2	M
25.	An observer 1.5 m tall stands 28.5 m away from a chimney. The angle of elevation of the top of the chimney from her eyes is 45° . Explain why we add the observer's height to the calculated height of the chimney when using trigonometric ratios to find the chimney's total height.	2	M
26.	Write the converse of the statement: 'If a triangle is equilateral, then all its interior angles are equal.' Also state whether the converse is true or false.	2	M
27.	A triangle ABC is similar to triangle DEF. The lengths of sides AB and DE are 6 cm and 9 cm respectively. If the perimeter of triangle DEF is 36 cm, find the perimeter of triangle ABC and explain the reasoning behind the calculation. OR Using elimination method, check whether the pair of equations $2x + 3y = 8$ and $4x + 6y = 7$ has a solution. Justify the result.	2	M
28.	A solid toy is a hemisphere surmounted by a right circular cone of height 2 cm and base diameter 4 cm. Find the volume of the toy and the difference between the volume of a cylinder circumscribing the toy and the toy's volume. (Take $\pi = 3.14$)	2	M
29.	A point P lies 7 cm from the centre O of a circle with radius 5 cm. How many tangents can be drawn to the circle from P, and what is the length of each tangent? OR A chord of a circle of radius 15 cm subtends an angle of 60° at the centre. Find the areas of the corresponding minor and major segments. Use $\pi = 3.14$ and $\sqrt{3} = 1.73$.	2	M

30.	A solid toy is made by placing a cone on top of a hemisphere such that their bases coincide. The diameter of the hemisphere and the base of the cone is 6 cm, and the height of the cone is 9 cm. Calculate the volume of the toy.	5 M
31.	A chord of a circle of radius 14 cm subtends an angle of 90° at the centre. Calculate the area of the corresponding segment of the circle. (Use $\pi = 22/7$) OR Two tangents TP and TQ are drawn to a circle with centre O from an external point T. If the chord PQ of the circle has length 10 cm and the radius of the circle is 6 cm, find the length of TP using properties of tangents and the related triangle similarity.	5 M
32.	Prove the trigonometric identity: $(\operatorname{cosec} \theta - \cot \theta)^2 = \cos \theta / (1 + \cos \theta)$ for an acute angle θ. OR If the points A(6, 1), B(8, 2), C(9, 4) and D(p, 3) are the vertices of a parallelogram taken in order, find the value of p using the property that diagonals of a parallelogram bisect each other.	5 M
33.	An ornithologist wants to estimate the number of parrots in a large field. She catches 32 parrots, rings them, and releases them back into the field. A week later, she catches 40 parrots, of which 8 are ringed. (i) What fraction of the second catch is ringed? (ii) Using this information, estimate the total number of parrots in the field. (iii) State the assumptions made in this model and explain why validating these assumptions is important. OR A blue die and a grey die are thrown simultaneously. The possible outcomes consist of ordered pairs where the first number is from the blue die and the second is from the grey die. There are 36 possible outcomes in total. (i) What is the probability that the sum of the numbers on the two dice is 8? (ii) What is the probability that the sum of the two dice is 13? (iii) What is the probability that the sum of the two dice is less than or equal to 12? (iv) Explain why the pair (1, 4) is considered different from (4, 1) in this experiment.	14 M

34.	A survey was conducted on the number of hours students of a class studied per week. The grouped frequency distribution of the data obtained is as follows: <table><tr><th>Class interval (hours)</th><th>Number of students (f_i)</th></tr><tr><td>0 - 5</td><td>4</td></tr><tr><td>5 - 10</td><td>10</td></tr><tr><td>10 - 15</td><td>15</td></tr><tr><td>15 - 20</td><td>8</td></tr><tr><td>20 - 25</td><td>3</td></tr></table> (i) Calculate the mean number of hours studied per week using the assumed mean method by taking the assumed mean as the class mark of the class 10-15. (ii) Verify your answer by applying the direct method. (iii) State one advantage of the assumed mean method over the direct method in this context. (iv) Interpret what the mean number of hours studied per week signifies for the class. OR Aftab tells his daughter, “Seven years ago, I was seven times as old as you were then. Also, three years from now, I shall be three times as old as you will be.” (i) Represent this situation algebraically by forming a pair of linear equations in two variables. (ii) Using the substitution method, find the present ages of Aftab and his daughter. (iii) Verify the solution by checking the given conditions. (iv) If Aftab's present age is denoted by s and his daughter's by t, write the values of s and t obtained.	Class interval (hours)	Number of students (f_i)	0 - 5	4	5 - 10	10	10 - 15	15	15 - 20	8	20 - 25	3	4 M
Class interval (hours)	Number of students (f_i)													
0 - 5	4													
5 - 10	10													
10 - 15	15													
15 - 20	8													
20 - 25	3													

SECTION B - VERY SHORT ANSWER

(1 x 2 = 2 Marks)

35.	A vertical pole 6 m high casts a shadow of 4 m. At the same time, a tower casts a shadow of 28 m. Find the height of the tower. OR Find the roots of the quadratic equation $x^2 + x - 6 = 0$.	2 M
-----	--	-----

SECTION C - SHORT ANSWER

(1 x 3 = 3 Marks)

36.	What is the probability of getting a sum of 7 when two dice are thrown together?	2 M
-----	--	-----

SECTION D - LONG ANSWER**(1 x 5 = 5 Marks)**

37.	Find the zeroes of the cubic polynomial $p(x) = 2x^3 - 5x^2 - 14x + 8$. Then verify the relationships between the zeroes and the coefficients of the polynomial.	5 M
-----	---	-----

SECTION E - CASE BASED QUESTIONS**(1 x 4 = 4 Marks)**

38.	Shakila puts money in her daughter's money box every year on her birthday starting with `100 on the first birthday, increasing the amount by `50 each subsequent birthday. (i) Calculate the total amount collected in the money box by her daughter's 10th birthday. (ii) What would be the amount put into the box on the 15th birthday? (iii) If Shakila stopped putting money after the 21st birthday, find the sum collected up to that point. (iv) On which birthday would the amount put into the money box reach `1000? OR A circle with centre O has a chord PQ of length 8 cm. The radius of the circle is 5 cm. Two tangents at points P and Q intersect at point T outside the circle. (i) Prove that triangle TPQ is isosceles. (ii) Find the length of the tangent segment TP. (iii) If OR is the perpendicular from O to chord PQ meeting PQ at R, find the length of segment PR. (iv) Explain why OT (where OT is the line joining centre O to point T) is perpendicular to PQ.	4 M
-----	--	-----